EG2A12 - Data Preparation Project report

ECO-WASTE

Group : A4

Team member:

Lim Hong Jean	250480M
Zi Qian	243339F
Kwan Teng	242308K
Pierre	252021M
Emmanuel	252252Z

Executive Summary

As Singapore’s sole offshore landfill, Pulau Semakau receives the waste across the nation to incinerate the waste. Given Singapore's acute land constraints and the environmental risks associated with waste disposal, reducing waste generation and elevating recycling rates are paramount to building a cleaner, more sustainable future.

To address this issue, our team sourced, cleaned, and processed data from reliable organizations to produce accurate datasets for analysis. Additionally, we benchmarked international waste management policies and practices to identify successful strategies that could be adapted for Singapore.

1. Business Understanding

1.1.1 Project Context

Singapore is a land-scarce country that relies on Semakau Landfill, the only landfill and the final stop for incineration ash and non-incinerable waste for the disposal of what cannot be recycled. At current disposal rate, the Ministry of Sustainability and the Environment (MSE) projects Semakau Landfill will run out of space by around 2035.

Under the Zero Waste Masterplan, Singapore has set a target of a 70% overall recycling rate and a 30% reduction in the daily per-capita waste sent to landfill, both by 2030

So, extending its lifespan depends on reducing the volume of waste sent there, which in turn depends on generating less waste in the first place and recycling more of what is generated.

This project investigates Singapore's waste generation and recycling trends using data-driven analysis, benchmarks Singapore's performance against international peers, and proposes solutions to help close the gap to these targets.

1.1.2 Problem Statement

"**How can Singapore improve its waste management and recycling efficiency?"**

1.1.3 Data Mining Goals

- Analyse the major contributors and trends in Singapore's waste generation by category, 2000-2024.
- Analyse Singapore's recycling-rate trends by waste type and identify how recycling performance has changed over time.
- Analyse Singapore's waste generated per capita and forecast future trends to 2034.
- Identify and rank OECD countries that achieved the greatest reduction in municipal waste generated per person from 2000-2020, to benchmark Singapore's waste-reduction performance.
- Identify countries with consistently high municipal recycling rates from 2000-2020 and recommend a transferable recycling solution for Singapore.

1.2 Data Sources & Dataset Exploration

Three publicly available datasets were selected to support the project's analysis of Singapore's waste generation, recycling performance and international benchmarking. Two datasets were obtained from data.gov.sg while the international comparison dataset was obtained from the Organisation for Economic Co-operation and Development (OECD). The use of both local and international sources allowed the analysis to examine Singapore's waste-management performance in its domestic context while also comparing it against other countries using standardised waste indicators.

The datasets were selected because no single source contained all the variables required to address the project's data-mining goals. Singapore's waste dataset provides information on the amount and type of waste generated and recycled, while the population dataset provides the denominator required to convert waste totals into a per-capita measure. The OECD dataset then provides internationally comparable waste-generation and recycling rates that can be used to identify the best performing country and establish benchmarks and policy that can be implemented in Singapore. Together, the three datasets support analysis at three levels: local waste performance, population-adjusted waste generation, and international comparison.

Dataset	Analytical Purpose	Key Variables	Relevance to Project
Singapore Waste Management & Recycling Statistics — data.gov.sg	Analyse local waste-generation and recycling patterns from 2000–2024.	Waste category, year, total waste generated, total waste recycled, overall recycling rate	Provides the main dataset for identifying major waste categories, changes in waste generation and Singapore's recycling performance over time.
Singapore annual Population — data.gov.sg	Adjust waste generation for changes in population size.	Year, total population	Allows total waste figures to be converted into waste generated per person, making it possible to determine whether changes in waste generation are driven mainly by population growth or by changes in individual waste-generation behaviour.

Oversea Countries Waste Management and Recycling Statistics	Benchmark Singapore against overseas countries and identify countries that achieved strong waste-reduction or recycling performance.	Country, year, municipal waste generated in kg/person, recycling rate (%)	Provides standardised international indicators that support cross-country comparison, country ranking and the selection of benchmark countries for policy analysis.

Singapore Waste Management & Recycling Statistics

The first dataset contains Singapore's annual waste-management and recycling statistics. Its key variables such as waste category, reporting year, total waste generated, total waste recycled and overall recycling rate. The project presentation identifies the available period as 2000–2024.

This dataset forms the foundation of the local analysis because it enables us to analyze waste performance from both a generation and recycling perspective. Waste-category data helps to figure out which materials contribute most to Singapore's overall waste stream, while the generated and recycled quantities can be used to examine how recycling performance has changed over time.

However, total waste generation alone may be misleading when analysing long-term trends because Singapore's population has also changed. A larger population may naturally generate a larger amount of total waste even if the amount generated by each individual remains unchanged or decreases. For this reason, the waste dataset was supplemented with annual population data.

Singapore Annual Population

The second dataset contains Singapore's annual population figures, with the key variables year and total population. The dataset covers the period from 2000–2025.

Its main analytical purpose is to allow waste generation to be normalised by population size. By combining population figures with annual waste-generation data, a waste-per-capita measure can be derived. This provides a more meaningful indicator of individual waste-generation behaviour than total waste tonnage alone.

For example, an increase in total waste does not necessarily mean that each resident is producing more waste; the increase may partly be explained by population growth. The per-capita transformation therefore helps separate the effect of population size from changes in waste-generation intensity and supports more meaningful year-to-year comparisons.

OECD Municipal Waste Statistics

The third dataset provides international waste-management statistics from the OECD. The available variables include country, reporting year, recycling rate and municipal waste generated in kilograms per person. The source dataset covers records from 1995–2020, while this project's benchmarking analysis focuses on the common 2000–2020 period.

The OECD dataset was selected because it provides standardised international indicators that allow waste-management performance to be compared across countries. This directly supports the project's data-mining objective of identifying and ranking countries that achieved the greatest reduction in municipal waste generated per person between 2000 and 2020.

Using kg/person rather than total national waste is particularly important for international comparison because countries differ significantly in population size. A per-capita indicator places countries on a more comparable basis and allows the analysis to focus on relative waste-generation performance rather than national scale.

The OECD data also supports the policy-benchmarking stage of the project. Countries demonstrating strong performance can be identified from the data, after which their waste-reduction or recycling approaches can be investigated as potential reference points for Singapore.

Overall Dataset Integration

The three datasets therefore serve complementary analytical roles. The Singapore waste dataset establishes what is happening to waste generation and recycling locally; the population dataset helps determine how much waste is being generated relative to the number of people; and the OECD dataset provides an external benchmark to evaluate how Singapore compares with other countries and what can potentially be learned from stronger performers.

This combination creates a more complete analytical foundation than any individual dataset could provide on its own and ensures that the subsequent data preparation, visualisation, forecasting and policy recommendations remain aligned with the project's business objective of improving Singapore's waste-management and recycling efficiency.

1.3 Data Exploration & Quality Issues

1.3.1 Missing Values

Several countries in the OECD dataset had incomplete yearly records. The table below, produced during initial exploration, summarises the extent of missing years for the countries most affected.Countries with substantial gaps (notably Canada and Costa Rica) were dropped from any analysis requiring a complete 2000-2020 time series, since interpolating two decades of missing data would introduce more error than it removes.

1.3.2 Inconsistent Data

The three datasets did not share a common structure out of the box: the OECD file used machine-readable column codes (e.g. REF_AREA, TIME_PERIOD, OBS_VALUE) rather than descriptive names, waste figures were reported in different units across datasets (tonnes for Singapore, kg per person for OECD), and year coverage differed between datasets (2000-2024 for Singapore, 1995-2020 for OECD). These had to be reconciled before the datasets could be joined or compared.

1.3.3 Unusual Fluctuations / Outliers

Some countries' year-on-year recycling rates showed unusually sharp swings. Sweden, for example, moved from a 29.9% recycling rate in 2018 to 32.5% in 2019 before dropping to 20.3% in 2020 — a fall of over 12 percentage points in a single year, most plausibly linked to pandemic-related disruption to waste collection and export markets in 2020 rather than a genuine change in Sweden's recycling capability. Because a swing of this size can distort a percentage-change calculation, 2020 was flagged and treated with caution in any single-year comparison, though it was retained in the full time series rather than removed outright, since excluding it would itself distort the historical trend.

1.4 Dataset Cleaning & Preparation

1.4.1 Standardising the Data

Column names across the OECD extract were renamed to descriptive, consistent labels — for example, “Reference area” and “TIME_PERIOD” became “country” and “year”, and the observation-value column was renamed “waste_per_person_kg” so that units were unambiguous at a glance.

The year field was cast to an integer type across all three datasets, and waste quantities were standardised to a common unit (kilograms) so that Singapore's tonnage figures and the OECD's per-person kilogram figures could be related to one another after the per-capita transformation described in section 1.5.

1.4.2 Handling Missing and Invalid Data

Missing yearly observations identified in OECD dataset during data exploration were excluded from analyses requiring complete 2000–2020 coverage. Large gaps, such as those found for Australia and New Zealand, were not interpolated to avoid introducing estimated values that could affect country comparisons.

1.4.3 Filtering Relevant Records

The OECD dataset was filtered to include only records relevant to the project's objectives. These included municipal waste, waste generated per person in kilograms, and recycling rate percentages. Countries with insufficient 2000–2020 coverage were excluded to ensure fair comparisons.

1.4.4 Duplicate and Invalid Record Checks

The datasets were checked for duplicate country-year and category-year records. No duplicates were found in the Singapore dataset.
In the OECD dataset, duplicate country-year records caused by different observation statuses were removed.

1.4.5 Data Validation

The cleaned datasets were validated to ensure that there were no missing values or duplicate keys in the fields used for analysis. Data types were also checked, with year stored as an integer and waste figures as numeric values. Waste and recycling values were checked against plausible ranges, and the year ranges were verified against the documented dataset coverage. These steps completed the Data Preparation phase of CRISP-DM and produced analysis-ready datasets.

1.5 Data Transformation & Derived Metrics

Following the cleaning and validation process, additional transformations were performed to convert the prepared datasets into meaningful analytical measures. These transformations were required because several of the project's data-mining objectives could not be answered directly using the raw variables. Derived measures such as waste generated per capita, percentage waste reduction and forecast scenario variables were therefore created to support comparison, trend analysis, international benchmarking and forecasting.

1.5.1 Waste Generated per Capita

Singapore's total waste generation alone does not provide a fair indication of changes in individual waste-generation behaviour because the country's population has changed considerably over time. Therefore, the Singapore waste dataset was combined with the annual population dataset using the year as the common field.

Waste generated per capita was calculated as:

Waste per Capita (kg/person) = (Waste Generated in Tonnes × 1,000) ÷ Population

For the municipal-waste comparison, selected household and commercial waste categories were first aggregated before being combined with Singapore's annual population figures. The resulting waste-per-capita measure expressed the amount of waste generated relative to the size of the population.

This transformation was necessary because an increase in total waste could partly result from population growth rather than an increase in the amount of waste generated by each person. Expressing waste in kg/person therefore enabled more meaningful comparisons across different years and also provided a common measurement unit for benchmarking Singapore against OECD countries.

1.5.2 OECD Waste Reduction Percentage

To identify the OECD countries that achieved the greatest long-term improvement in municipal waste generation, each country's waste generated per person in 2000 was compared with its corresponding value in 2020.

The overall reduction percentage was calculated as:

Waste Reduction (%) = [(Waste in 2000 − Waste in 2020) ÷ Waste in 2000] × 100

The cleaned data was first reshaped so that each country's 2000 and 2020 waste-per-person values could be compared directly. The absolute kilogram difference and percentage change were then calculated, after which the countries were ranked from the largest reduction to the smallest.

A percentage measure was selected instead of relying only on the absolute change in kilograms. This allowed improvement to be assessed relative to each country's original level of waste generation and provided a more comparable basis for international benchmarking. The resulting metric directly supported the project's data-mining goal of identifying the countries that achieved the greatest reduction in municipal waste generated per person from 2000 to 2020.

1.5.3 Year-to-Year Trend Measures

Additional measures were created to analyse the pattern of change within the 2000–2020 period rather than relying only on the two endpoint years.

For each country, the analysis calculated the annual kilogram difference and percentage change in waste generated per person. Each year was then classified as an Increase, Decrease or No Change compared with the previous year.

The number of increasing and decreasing years was subsequently summarised for each country.

This transformation provided additional context for the country ranking. A country may record a lower value in 2020 than in 2000 even if its waste levels fluctuated considerably between these years. Examining the yearly direction of change therefore helped determine whether the overall reduction reflected a more sustained downward trend and supported the later investigation of policies implemented by the highest-performing countries.

1.5.4 Forecasting and Policy Scenario Dataset

The historical Singapore waste-per-capita data was also transformed into a structure suitable for future scenario analysis. The existing baseline forecast was retained as the Forecast Without Policy scenario. A second scenario, Forecast with Policy, was then created using the average annual reduction rate derived from the selected OECD benchmark countries.

The forecast dataset included additional variables for:

- forecast waste per person without the benchmark policy effect;
- forecast waste per person with the benchmark reduction rate;
- years since the policy scenario began;
- waste prevented in kg/person; and
- percentage reduction between the two forecast scenarios.

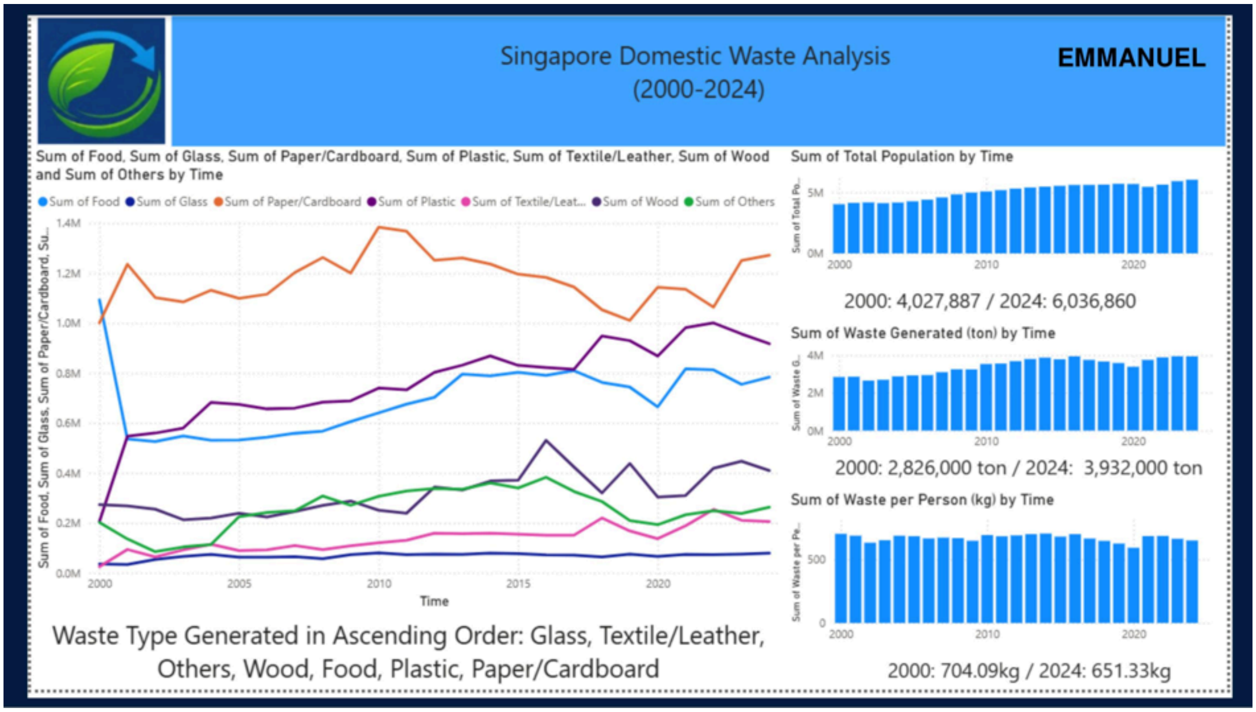
The benchmark reduction was applied cumulatively across the forecast period rather than as a one-time decrease. The difference between the baseline and policy scenarios was then calculated to estimate the amount of waste potentially prevented.

This transformation allowed the historical trend and both future scenarios to be visualised together in Power BI. Its purpose was not to predict with certainty that the proposed recommendation would achieve the same results as the benchmark countries. Instead, it provided a benchmark scenario showing the potential outcome if Singapore were able to achieve a similar annual waste-reduction rate.

Overall, these transformations converted the cleaned datasets into analytical measures directly aligned with the project's data-mining objectives. They enabled fairer population-adjusted comparisons, international waste-reduction ranking, examination of year-to-year trends and evaluation of a future waste-reduction scenario for Singapore. These activities form part of the Data Preparation phase of CRISP-DM, where cleaned data is transformed into variables suitable for subsequent visualisation and analysis.

2. Data Visualization, Insights & Recommendations

2.1 Singapore Waste Generation Analysis



Visualization:

A multi-series line chart was used to track seven major waste categories—Food, Glass, Paper/Cardboard, Plastic, Textile/Leather, Wood, and Others—from 2000 to 2024. Supporting bar charts were also included to illustrate Singapore's total population and waste generated per person over the same period. A line chart was selected because it effectively demonstrates changes and long-term trends across multiple waste categories, allowing comparisons to be made over the 25-year period.

Findings:

Paper and cardboard consistently accounted for the largest proportion of domestic waste throughout the period, averaging approximately 1.2 million tonnes annually. In comparison, glass remained the smallest contributor, with annual waste generation amounting to only a few thousand tonnes. Plastic and food waste also demonstrated an overall upward trend from 2000 onwards and have become two of the major waste categories, following paper and cardboard.

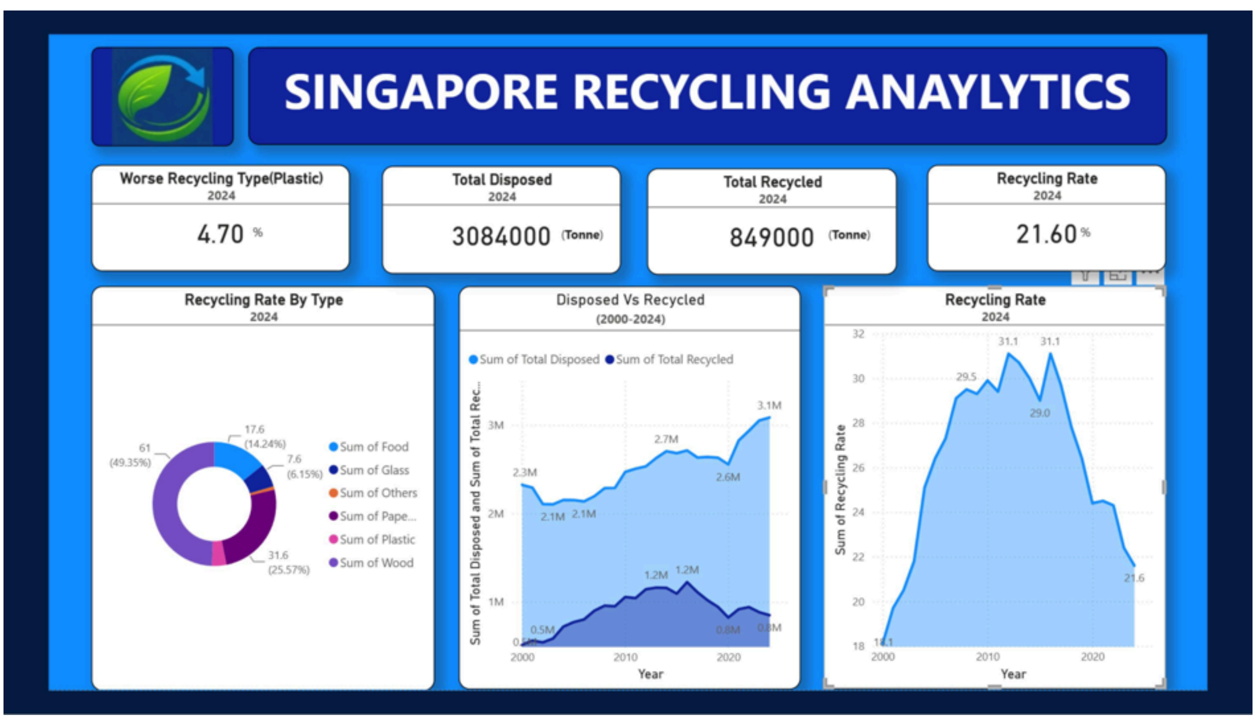
Interpretation:

Singapore's population increased from 4,027,887 in 2000 to 6,036,860 in 2024, representing an increase of approximately 50%. Over the same period, total waste generated increased from 2.83 million tonnes to 3.93 million tonnes. However, waste generated per person decreased slightly from 704.09 kg in 2000 to 651.33 kg in 2024. This suggests that although Singapore's total waste generation has increased alongside population growth, the amount of waste generated per individual has declined. This indicates a partial decoupling between population growth and per-capita waste generation.

Implication:

Since paper/cardboard, plastic, and food waste constitute a significant proportion of Singapore's domestic waste, prioritising these three categories could provide greater opportunities for reducing overall waste generation. Targeted measures such as improving recycling practices, reducing unnecessary consumption, and increasing public awareness of proper waste disposal could therefore be more effective than applying broad, category-independent waste management strategies.

2.2 Singapore Recycling Performance Analysis



2.2.1 Overall Recycling Trend

The recycling-rate trend derived from the dataset shows that Singapore's domestic waste recycling rate increased from approximately 18% in 2000 to a peak of around 31% in 2018–2019, before declining to approximately 21.6% in 2024. This represents a decrease of nearly 10 percentage points over approximately five to six years. Although the magnitude differs, this trend is broadly consistent with the overall recycling-rate figures reported by the National Environment Agency (NEA). NEA's published overall recycling rate remained close to 60% from 2013 before declining to 57% in 2022, 52% in 2023, and 50% in 2024. The domestic-sector recycling rate also declined from approximately 20% before 2020 to around 11–12% between 2023 and 2025.

It is important to note that the recycling rates calculated in this project are based solely on the domestic waste-category dataset and therefore do not represent NEA's full national recycling-rate calculation. NEA's overall figures incorporate additional waste streams, including construction and demolition waste and non-domestic waste, which are accounted for differently. Therefore, the two datasets should not be compared directly on a year-by-year basis. Nevertheless, both datasets indicate a similar underlying trend: Singapore's recycling performance has declined since approximately 2019–2020. This decline has been attributed by the Ministry of Sustainability and the Environment (MSE) to factors including increased freight costs and weaker overseas demand for recyclable materials following the COVID-19 pandemic.

2.2.2 Recycled vs. Disposed Waste

Total waste disposed increased substantially from approximately 2.6 million tonnes to 3.1 million tonnes in recent years, while the total amount of waste recycled remained relatively stable at approximately 0.8 million tonnes. This indicates that the increase in waste generation has not been matched by a corresponding increase in the volume of waste recycled. The finding is consistent with NEA's reported trends, which indicate that waste generation has continued to increase while recycling volumes have not increased at the same pace. If this trend continues, the growing volume of waste requiring disposal could place additional pressure on Singapore's limited landfill capacity and contribute to a reduction in the remaining lifespan of Semakau Landfill.

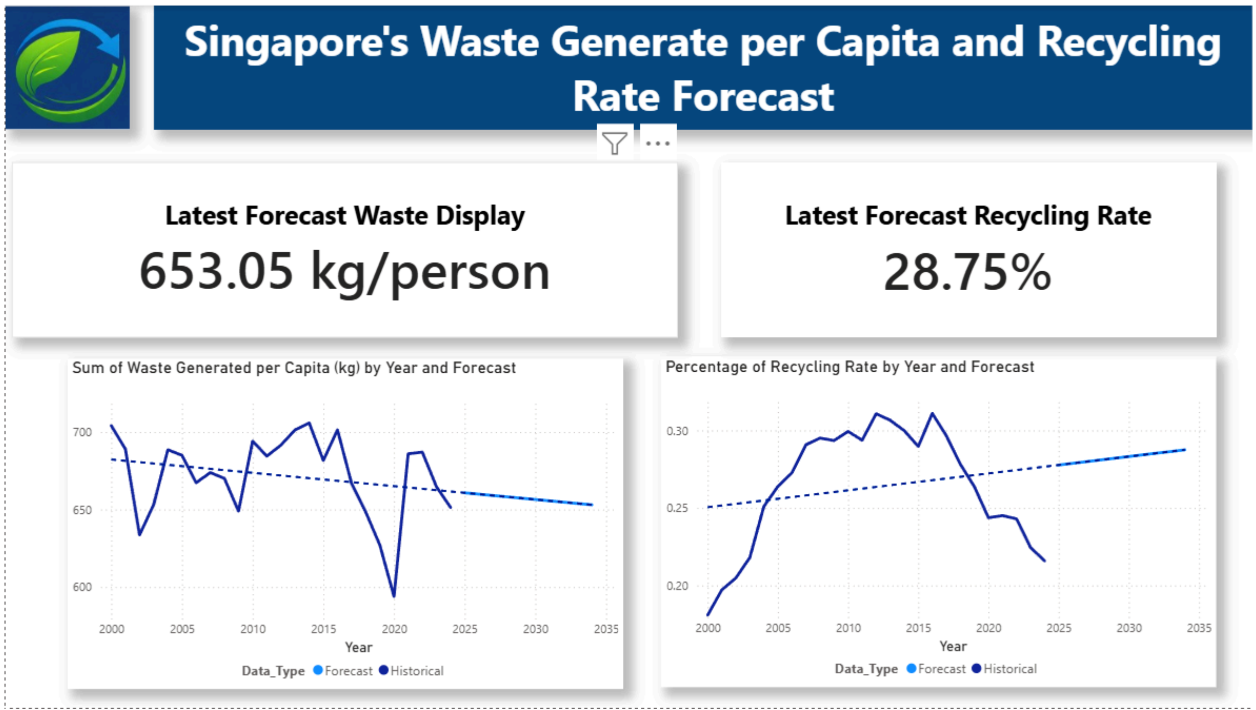
2.2.3 Plastic Recycling

Plastic is one of the major waste categories generated in Singapore; however, it has the lowest recycling rate among the categories analysed in the dashboard, at 4.70%. This finding is broadly consistent with figures reported separately by MSE, which place Singapore's plastic recycling rate at approximately 4%.

The substantial gap between the volume of plastic waste generated and the amount successfully recycled highlights several challenges within Singapore's recycling system. One contributing factor is contamination, as NEA and MSE have noted that a significant proportion of materials placed in recycling bins may be rejected because they are contaminated or unsuitable for recycling. In addition, changes in international recycling markets and tighter restrictions on recyclable-waste imports by major importing countries have further affected the recovery of plastic waste.

Therefore, plastic waste represents a key priority area for improvement. Increasing plastic recovery and reducing contamination could help improve Singapore's overall recycling performance and reduce the amount of plastic waste sent for disposal. This issue is directly addressed by the Return & Save recommendation presented in Section 2.6.

2.3 Singapore Waste & Recycling Forecast



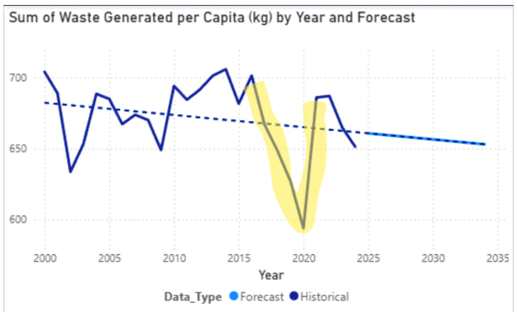
2.3.1 Forecasting Method

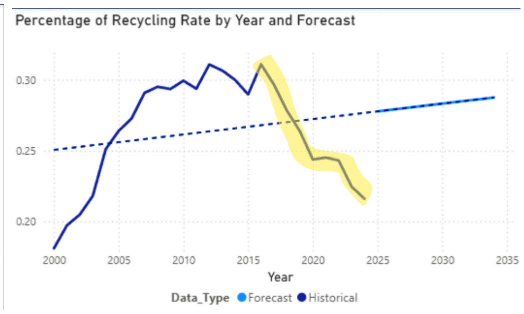
A trend-based linear regression model was developed in Power BI using Singapore's historical Waste Generated per Capita and Recycling Rate data from 2000 to 2024. The model was then used to project both indicators through 2034. Linear regression was selected because it provides a transparent and straightforward approach for identifying long-term trends and generating forecasts. This approach was considered appropriate given the relatively limited historical dataset and the significant disruption caused by the COVID-19 pandemic during the observed period.

2.3.2 Forecast Results

The latest forecast values indicate 653.05 kg per person for waste generated per capita and a 28.75% recycling rate. The forecast for waste generated per capita shows an overall downward trend, although the trend is influenced by the significant decline observed in 2020. In contrast, the recycling rate shows an overall upward trend in the forecast despite the decline observed in recent years. This is because the linear regression model considers the entire 2000–2024 historical period, during which the recycling rate generally demonstrated an upward trend over the longer term.

2.3.3 Impact of COVID-19





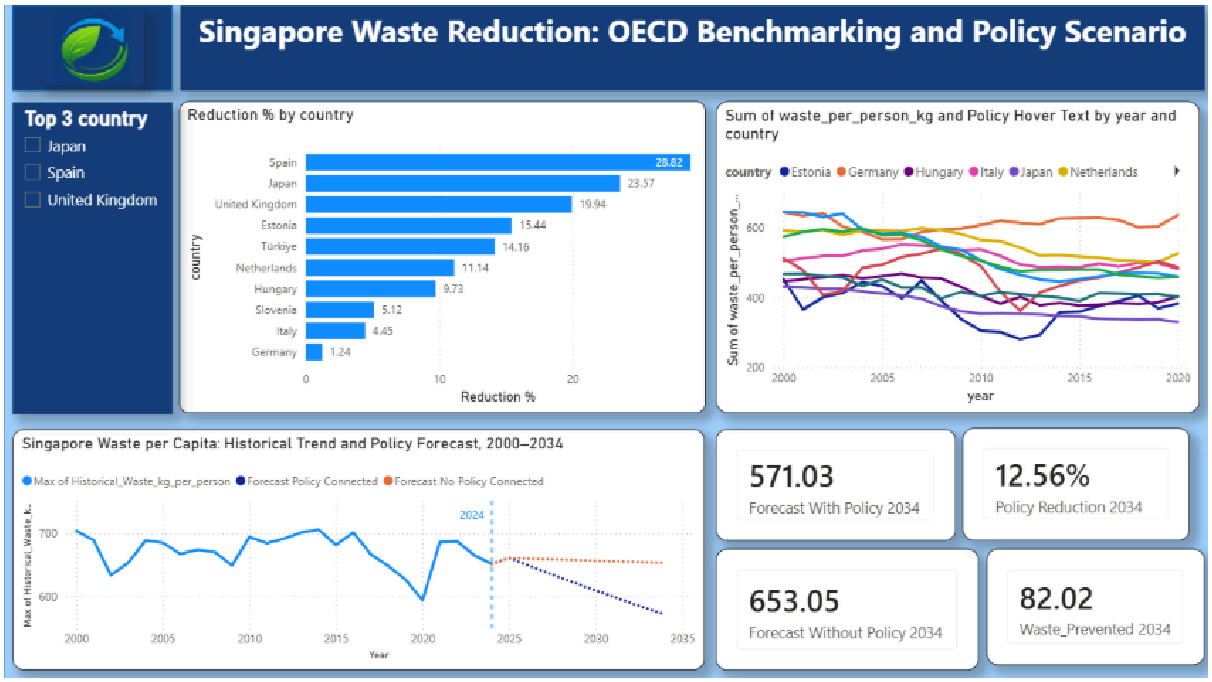
Both indicators exhibit a noticeable deviation around 2020, reflecting the impact of the COVID-19 pandemic. Waste generated per capita declined significantly during this period, primarily due to reduced economic, commercial, and industrial activities during Singapore's Circuit Breaker period, before subsequently recovering. The recycling rate also declined during this period, partly due to disruptions in freight transportation and overseas recycling markets, as discussed in Section 2.2.1. However, unlike waste generated per capita, the recycling rate has not yet fully returned to its pre-pandemic level.

2.3.4 Forecast Limitations

- The linear regression model is based on 25 years of historical data and includes the significant disruption observed in 2020. As a simple trend-based model, it cannot distinguish between temporary fluctuations and permanent structural changes in the underlying trend.
- The forecast assumes that historical trends will continue into the future and does not account for potential changes resulting from new government policies, technological developments, or waste management initiatives, including the recommendations proposed in Section 2.6.
- External factors, such as global commodity prices for recyclable materials, freight and transportation costs, and overseas restrictions on recyclable waste imports,

are not incorporated into the model. Changes in these factors could significantly influence Singapore's future waste generation and recycling rate.

2.4 OECD Waste Reduction Benchmarking and Policy Scenario



2.4.1 Benchmarking Method

To identify international examples of strong waste-reduction performance, OECD countries were compared based on the change in municipal waste generated per person between 2000 and 2020. The percentage-reduction metric derived in Section 1.5 was used so that countries could be assessed relative to their own starting levels rather than solely by the absolute number of kilograms reduced.

Only countries with sufficiently complete annual observations across the 2000–2020 analysis period were retained. This was important because comparing countries with substantially different levels of data completeness could distort the ranking. Using a common time period, common municipal-waste definition and common unit of kg/person created a more consistent basis for international comparison.

The countries were ranked from the largest percentage reduction to the smallest. The purpose of this ranking was not only to identify high-performing countries, but also to select suitable benchmarks for the subsequent policy analysis. The highest-performing countries were therefore examined further to determine whether their waste-management approaches contained lessons that could be adapted to the Singapore context.

2.4.2 Ranking Results

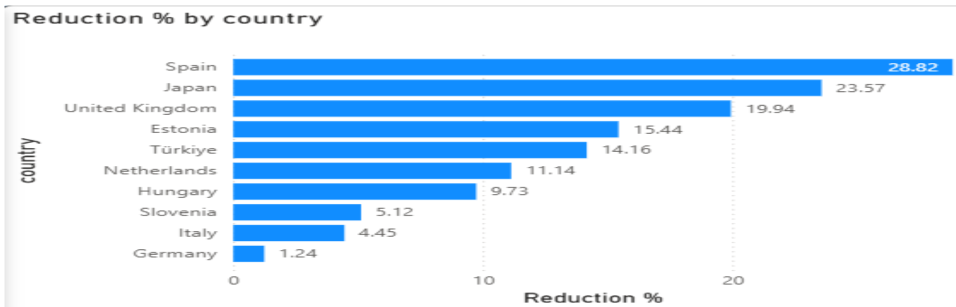
Rank	Country	Reduction 2000-2020
1	**Spain**	**28.82%**
2	**Japan**	**23.57%**
3	**United Kingdom**	**19.94%**
4	Estonia	15.44%
5	Türkiye	14.16%
6	Netherlands	11.14%
7	Hungary	9.73%
8	Slovenia	5.12%
9	Italy	4.45%
10	Germany	1.24%

The results show a clear separation between the three strongest performers and most of the remaining countries. Spain recorded the largest reduction at 28.82%, followed by Japan at 23.57% and the United Kingdom at 19.94%. These countries were therefore selected as the principal benchmarks for deeper policy investigation.

2.4.3 Visualisation Rationale

A horizontal bar chart was selected to present the country ranking because the main analytical purpose is comparison rather than time-series analysis. Sorting the bars from the largest to the smallest reduction allows the strongest-performing countries to be identified immediately.

The horizontal orientation was also more appropriate than a vertical column chart because several country names are relatively long. Horizontal labels improve readability while allowing the percentage differences between countries to remain visually clear.



The ranking visual therefore provides both an overview of international performance and a clear justification for selecting Spain, Japan and the United Kingdom for the next stage of analysis.

2.4.4 Interpretation of Benchmarking Results

The ranking indicates that substantial reductions in waste generated per person have been achieved in several OECD countries, although the scale of improvement varies significantly. Spain, Japan and the United Kingdom were particularly important because they achieved the three largest reductions while using different approaches to waste management.

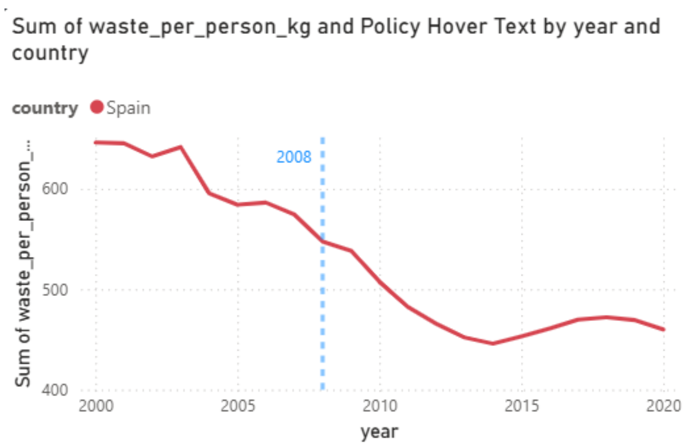
However, the ranking itself does not establish why these reductions occurred. A declining waste trend may be influenced by multiple factors, including policy interventions, economic conditions, changes in consumption patterns, technology and other structural changes.

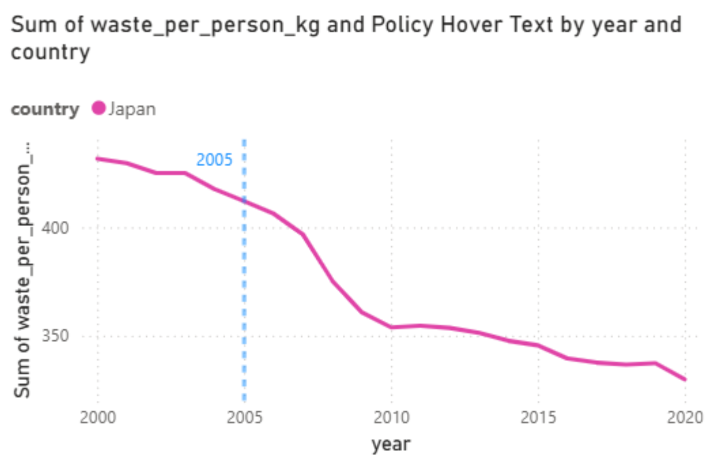
For this reason, the benchmarking results were used as a starting point for policy investigation, rather than as proof of policy causation. The next stage examined significant waste-related policies introduced in the three highest-performing countries and identified the mechanisms that could potentially be transferred to Singapore.

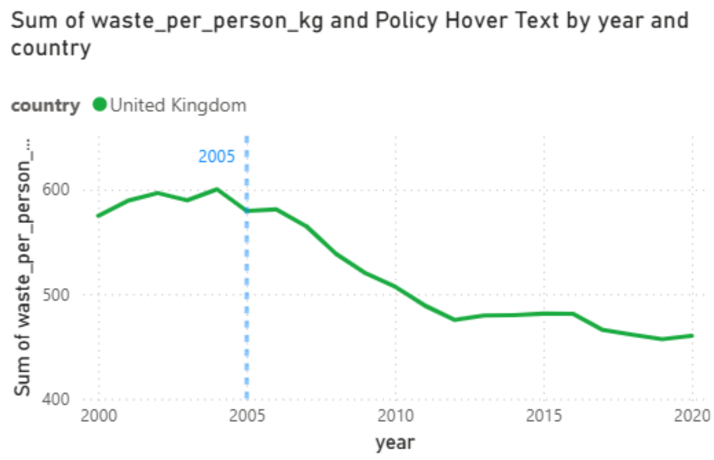
2.4.5 International Policy Analysis

Following the quantitative ranking, the three highest-performing benchmark countries were examined qualitatively to identify major waste-management interventions implemented during their respective declining waste trends.

The project adopts the assumption that the selected policy for each country represents a key contributing intervention associated with the country's improvement, while recognising that it cannot be treated as the sole cause of the entire reduction. This is consistent with the project's stated policy assumption.







Country	Selected Policy	Main Mechanism	Transferable Lesson for Singapore
Spain	Single-Use Shopping Bag Reduction Measures, 2008	Waste prevention at source	Reduce unnecessary disposable products before they enter the waste stream
Japan	Waste Collection Charging Policy, 2005	Financial incentive	Give individuals a direct economic reason to change waste-related behaviour
United Kingdom	Courtauld Commitment — Phase 1, 2005	Business participation and shared responsibility	Involve retailers and manufacturers directly rather than relying only on consumers

2.4.6 Spain — Waste Prevention at Source

Spain's selected intervention focused on reducing the use of single-use shopping bags. The measures targeted a 50% reduction in single-use shopping bags and involved working with retailers to reduce bag distribution while encouraging reusable alternatives.

The main lesson derived from Spain is the importance of preventing waste before it is generated. Instead of relying only on recycling or disposal after a product has already become waste, source-reduction measures attempt to reduce the quantity of unnecessary disposable material entering the waste stream in the first place.

For Singapore, this suggests that waste-management strategies should consider opportunities to replace frequently discarded single-use products with reusable alternatives.

2.4.7 Japan — Financial Incentives for Behaviour Change

Japan's selected policy involved encouraging municipalities to introduce waste-collection charging mechanisms. The underlying principle is that when waste disposal has a visible financial cost, households have a stronger incentive to reduce the amount of waste they generate.

The transferable lesson is not necessarily that Singapore should directly copy Japan's waste-charging structure. Instead, Japan demonstrates the broader principle that financial incentives can influence waste-related behaviour.

For Singapore, this principle can be adapted in a more positive form by using a refundable deposit. Rather than financially penalising consumers for generating waste, consumers can be encouraged to perform a desired action by receiving their money back when they return a reusable item.

2.4.8 United Kingdom — Business Participation

The United Kingdom's Courtauld Commitment — Phase 1 involved major supermarkets, food companies and manufacturers working together to reduce household food waste and unnecessary packaging.

The important lesson is that waste reduction cannot depend entirely on individual consumer behaviour. Businesses influence the type of packaging placed on the market, the availability of reusable alternatives and the infrastructure available to consumers.

Therefore, an effective Singapore waste-reduction strategy should involve businesses as active participants rather than placing the full responsibility on households.

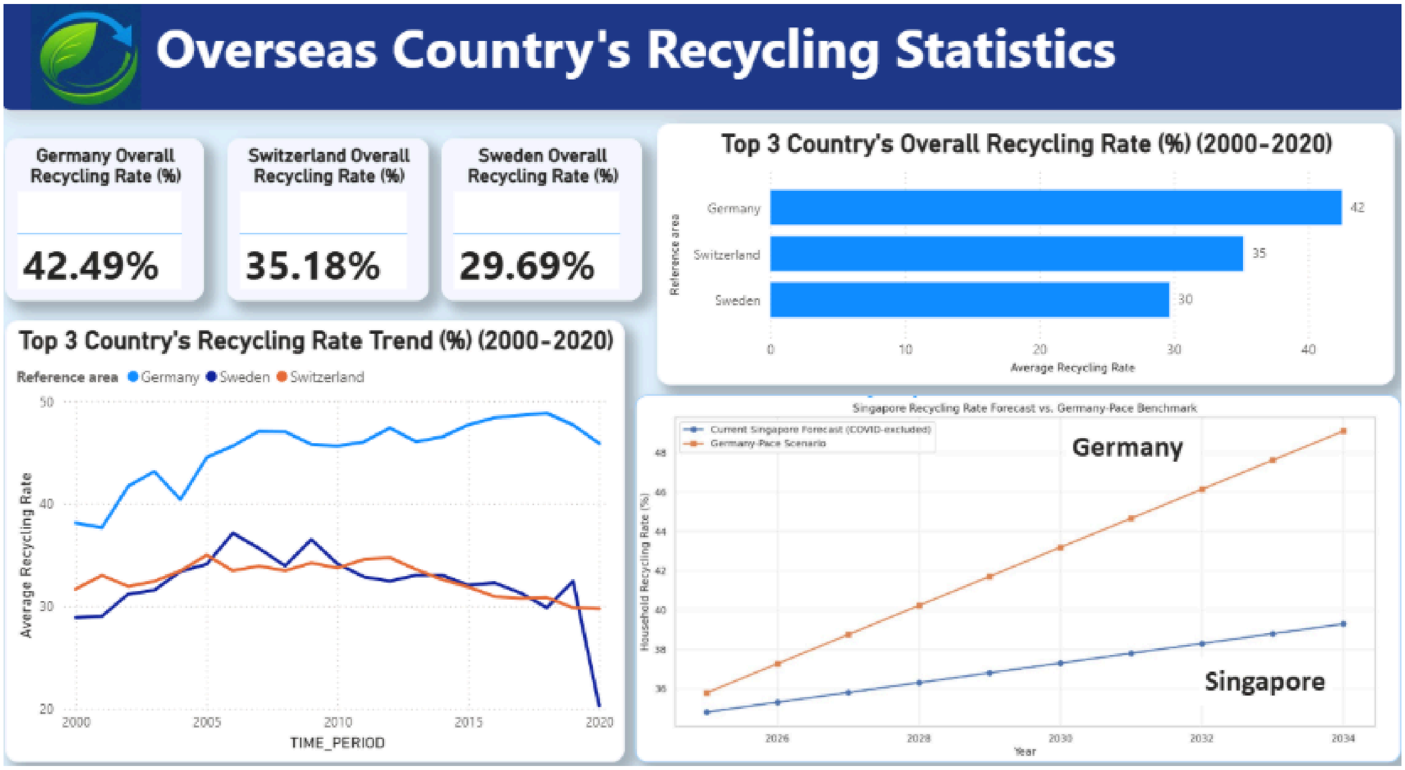
2.4.9 Combined Policy Lessons

Although the three benchmark countries used different approaches, their policies provide three complementary principles:

Spain → Prevent unnecessary single-use waste
Japan → Use financial incentives to motivate behaviour
United Kingdom → Involve businesses in shared responsibility

These lessons were not copied directly into the Singapore context. Instead, they were combined to develop a recommendation that addresses a specific preventable waste stream while considering the behaviour of both consumers and businesses.

2.5 OECD Overseas Country's Recycling Rate Statistic



2.5.1 Ranking method

The average municipal waste recycling rate (%) for each OECD country was calculated over the 2000–2020 period. Countries were then ranked in descending order, with the highest average recycling rate receiving the highest ranking.

2.5.2 Top 3 Countries based on overall recycling rate

Rank	Country	Recycling rate (%) from 2000-2020
1	Germany	42.49%
2	Switzerland	35.18%

3	Sweden	29.69%

2.5.3 Visualization Rationale

KPI Cards: Used to highlight the overall average recycling rate of the top three performing countries, allowing users to quickly identify the leading countries and their recycling performance.

Bar Chart: Used to rank the top three OECD countries by their average municipal waste recycling rate from 2000–2020. The horizontal format makes differences between countries easy to compare.

Line Chart: Used to show the recycling rate trends of Germany, Switzerland, and Sweden over 2000–2020. This allows changes in recycling performance over time and differences between countries to be identified.

Forecast Line Chart: Used to compare Singapore's projected recycling rate with the Germany-based benchmark scenario from 2025–2034. This helps illustrate the potential improvement Singapore could achieve if a similar performance trajectory to Germany were adopted

2.5.4 Interpretation

Germany recorded the highest average recycling rate (42.49%) from 2000–2020, followed by Switzerland (35.18%) and Sweden (29.69%). From the time trend graph, Germany also demonstrated a generally increasing recycling trend while compared to Switzerland and Sweden inconsistent recycling rate, making Germany a suitable benchmark for Singapore. The forecast indicates that Singapore's projected recycling rate remains below the Germany-based benchmark, highlighting potential opportunities for further improvement.

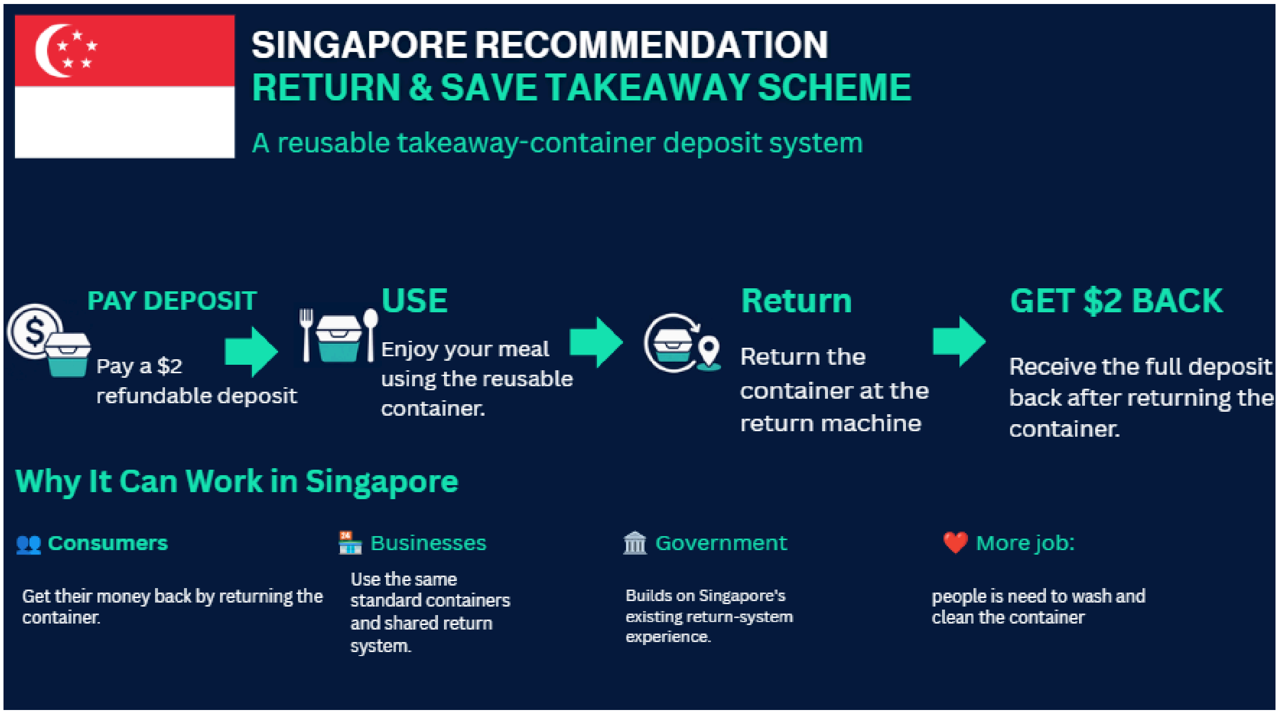
2.5.5 Germany policy analysis

Germany's recycling performance can be linked to its producer responsibility and separate collection policies, supported by the Packaging Ordinance introduced in 1991.

Policy	Policy Direction	Main Mechanism	Potential Lesson for Singapore
Producer Responsibility	Shift responsibility for packaging waste towards producers	Producers are made responsible for the collection, recovery and recycling of packaging waste	Strengthen producer responsibility to encourage companies to reduce packaging waste and improve recyclability
Separate Collection	Promote source separation of recyclable materials	Recyclable packaging is collected separately from general household waste through systems such as the **Yellow Bag**	Adapt source separation to reduce contamination and the rejection of recyclable materials, thereby improving Singapore's recycling rate

3. Recommendations

3.1 Recommendation 1 — Return & Save Takeaway Scheme



3.1 Recommendation 1 — Return & Save Takeaway Scheme

Strategic Opportunity: Prevent Waste Before Recycling Is Required

The analysis identifies a strategic weakness in Singapore's current waste-management performance: although waste generated per person has declined slightly over time, Singapore continues to generate a substantial volume of waste, while several important waste streams remain difficult to recover effectively. Plastic is particularly significant because it is one of the major waste categories identified in the dashboard but recorded the lowest recycling rate among the analysed categories, at 4.70%.

This suggests that relying only on downstream recycling may be insufficient for materials with consistently low recovery rates. A complementary strategy is therefore to reduce the amount of disposable material entering the waste stream in the first place.

Based on this insight and the subsequent OECD policy benchmarking, the group proposes the Return & Save Takeaway Scheme, a reusable takeaway-container deposit system based on waste prevention through reuse.

The recommendation combines three transferable principles identified from the three strongest waste-reduction benchmark countries:

Spain — Prevent waste at source: reduce unnecessary single-use products before they become waste.

Japan — Use financial incentives: provide a direct economic reason for consumers to perform the desired waste-reduction behaviour.

United Kingdom — Share responsibility with businesses: involve retailers and businesses in delivering the waste-reduction system rather than relying solely on individual consumers.

Rather than directly copying any one overseas policy, these principles are combined and adapted to Singapore's takeaway-food environment. The proposed system therefore addresses both the consumer behaviour and business participation required for a reusable-container system to operate effectively.

Proposed Operating Model

The proposed system follows a closed-loop reuse process:

Deposit → Use → Return → Refund → Clean → Reuse

When purchasing takeaway food from a participating merchant, customers may receive their meal in a standardised reusable food container instead of conventional single-use packaging. A S$2 refundable deposit is collected when the container is issued.

After consuming the meal, the customer returns the container at a participating return point or return machine and receives the full S$2 deposit back. Returned containers are then collected, professionally washed, inspected for damage and redistributed to participating merchants for subsequent use.

The S$2 deposit is proposed as an initial pilot amount rather than a permanently fixed value. Its effectiveness should be evaluated using actual return behaviour and adjusted if the amount does not provide a sufficient incentive for consumers to return the containers.

3.1.1 Consumer Participation

The refundable deposit provides consumers with an immediate financial reason to return the container. Instead of depending entirely on environmental awareness or voluntary behaviour, the system connects the desired behaviour with a direct economic incentive.

Consumers who return the container recover the full deposit and therefore do not incur an additional cost. In contrast, consumers who keep or fail to return the container forfeit the deposit.

This adapts the key lesson identified from Japan. However, rather than directly charging consumers according to the amount of waste they dispose of, the proposed scheme uses the financial mechanism in a positive incentive structure, where the desired action is rewarded through a full refund.

Financial motivation alone, however, may not be sufficient. Convenience is likely to be a major determinant of participation. Return points should therefore be located in places that consumers already visit frequently, such as hawker centres, food courts, shopping malls and other participating food-service locations. A convenient return network reduces the effort required from consumers and increases the likelihood that containers remain within the reuse system.

3.1.2 Business Participation

Businesses form the second major component of the scheme. Participating food retailers would use a common reusable-container standard and shared return infrastructure, rather than each business developing its own separate container system.

This approach reflects the lesson identified from the United Kingdom: effective waste reduction requires participation from businesses as well as consumers.

A shared system would also allow customers to return containers at participating return points without necessarily returning them to the original merchant. This improves convenience for consumers while creating a larger common pool of reusable containers.

To minimise the operational burden on individual food outlets, activities such as collection, washing, quality inspection, tracking and redistribution could be managed through a centralised service provider or shared operating system. Individual merchants would therefore not be required to develop their own washing and reverse-logistics infrastructure.

The collection, cleaning and redistribution process may also create additional operational demand for roles associated with container handling, hygiene inspection and logistics, although this would be a secondary benefit rather than the main objective of the scheme.

3.1.3 Government and System-Level Role

Government support would be important during the establishment of the scheme, particularly in setting common operating requirements, hygiene standards, container specifications and rules governing participating merchants and return locations.

The scheme could also build on Singapore's existing experience with return-system concepts, reducing the need to introduce a completely unfamiliar behavioural model.

However, the proposed system should not initially be implemented nationwide. A controlled pilot would allow the government, participating businesses and system operators to evaluate consumer behaviour, operational requirements and costs before committing to wider deployment.

3.1.4 Expected Waste-Reduction Mechanism

The main expected effect of the Return & Save Scheme is a reduction in the number of single-use takeaway containers entering Singapore's waste stream.

Under the conventional takeaway model, a food container typically becomes waste after a single transaction. Under the proposed system, one reusable container can support multiple takeaway transactions before eventually reaching the end of its usable life.

The waste-reduction mechanism can therefore be represented as:

Fewer single-use containers required per transaction → fewer containers discarded → lower packaging waste generated

This directly applies the waste-prevention principle identified from Spain. Instead of focusing only on what happens after packaging is discarded, the scheme reduces the need to create and dispose of a new single-use container for every takeaway meal.

Importantly, the recommendation should not be classified primarily as a recycling initiative. Its intended environmental benefit comes from extending the useful life of the same container through repeated reuse and thereby reducing the amount of disposable packaging required.

The international policy evidence provides the conceptual basis for the recommendation but should not be interpreted as proof that the selected policies alone caused the full waste reduction observed in Spain, Japan or the United Kingdom. The project assumes that the selected interventions were key contributing factors associated with the observed improvements, while recognising that economic conditions, other policies, technology and changes in consumer behaviour may also have influenced the long-term trends.

3.1.5 Potential Limitations and Mitigation

Several practical risks would need to be addressed before large-scale implementation.

Container return rate.

The system requires a sufficiently high proportion of containers to be returned. If customers frequently keep or discard them, replacement containers would have to be manufactured and purchased, reducing both the environmental and economic benefits of reuse.

Mitigation: Monitor the return rate during the pilot and adjust the deposit amount if S$2 does not provide a sufficient incentive.

Return-point convenience.
Consumers may be unwilling to participate if returning a container requires significant additional travel or effort.

Mitigation: Locate return points at high-footfall areas and participating food-service locations, and evaluate whether additional return points improve participation.

Hygiene and food safety.
Containers are used repeatedly for food and must therefore meet appropriate hygiene standards throughout their usable life.

Mitigation: Establish a controlled professional cleaning and inspection process. Containers that are cracked, damaged or otherwise unsuitable for reuse should be removed from circulation.

Infrastructure and operating costs.
The scheme would require investment in containers, return infrastructure, collection, washing, tracking and redistribution.

Mitigation: Begin with a limited pilot to evaluate operating costs and identify the most efficient logistics model before wider expansion.

Business participation.
A small number of participating merchants would limit the convenience and usefulness of a shared return system.

Mitigation: Introduce the scheme in concentrated locations where multiple food businesses can participate within the same return network before considering broader rollout.

3.1.6 Pilot Implementation and Evaluation

The group therefore recommends introducing the Return & Save Scheme initially as a controlled pilot programme in selected high-footfall locations, such as hawker centres, food courts or shopping malls.

A pilot allows the proposed system to be tested under real operating conditions before nationwide implementation. In particular, it would provide evidence on whether the proposed deposit is sufficient, whether consumers find the return process convenient and whether participating businesses can integrate the system without excessive operational burden.

The pilot should monitor the following key performance indicators:

Evaluation Area	**Suggested KPI**	**Purpose**
Consumer behaviour	Container return rate (%)	Measures whether the deposit successfully encourages returns
Waste prevention	Number of single-use containers avoided	Measures the direct waste-prevention benefit
Container management	Container loss rate (%)	Identifies how many reusable containers leave the system
Public participation	Number / percentage of customers using the scheme	Measures consumer acceptance

Business participation	Number of participating merchants	Evaluates scalability and business adoption
Operational performance	Average cost per reuse cycle	Assesses financial feasibility
Convenience	Average distance / availability of return points	Evaluates accessibility of the return network
Hygiene and quality	Number of rejected or damaged containers	Monitors food-safety and container-quality requirements

The results should then be used to refine the deposit amount, container design, return-point density, washing capacity, logistics arrangement and merchant participation model before any wider rollout.

Overall Recommendation

The Return & Save Takeaway Scheme translates the international benchmarking findings into a practical Singapore-focused intervention by combining waste prevention, financial motivation and shared business responsibility.

Its objective is not to manage disposable packaging more effectively after it becomes waste, but to reduce the need for disposable packaging in the first place.

By creating a closed-loop system in which containers are issued, returned, professionally cleaned and reused, the proposed scheme has the potential to reduce single-use takeaway waste while distributing responsibility across consumers, businesses and system operators.

Nevertheless, its effectiveness should be demonstrated through a controlled pilot rather than assumed. Expansion should only be considered after measurable evidence shows that the system can achieve sufficiently high return rates, meaningful reductions in single-use packaging and acceptable operating costs.

3.2 Recommendation 2 - Germany’s Waste separation adoption

Based on the analysis of Germany's recycling policies, a source-separation recycling system is recommended for Singapore. The proposed solution adapts Germany's Yellow Bag concept to help Singapore increase its recycling rate.

3.2.1 Solution Breakdown

Dedicated yellow bags for each household :

Each household will be provided with dedicated yellow bags for storing clean and recyclable materials, separating them away from the general household waste.

Convenient collection :

Residents can leave their filled yellow bags outside their doors on designated collection days, where collection workers will collect them.

This provides a convenient recycling method for residents while creating additional job opportunities for workers responsible for collecting the bags.

Contamination Control :

Collection workers will conduct visual checks to ensure that the bags contain suitable recyclable materials before collection.
This helps prevent contaminated materials from entering the recycling stream and being rejected during processing

Reward System :

Households that consistently meet the required recycling participation or quality criteria will receive vouchers as an incentive.
This encourages residents to participate regularly and practise proper waste separation.

3.2.2 Consumer Participation :

The proposed system provides households with dedicated yellow bags for collecting clean and recyclable materials. By providing a specific bag for recyclables, the system gives residents a clear and simple method of separating recyclable materials from general household waste.

Consumer participation is supported through both convenience and financial incentives. Instead of requiring residents to travel to a separate recycling facility or collection point, households can simply leave their filled yellow bags outside their doors on designated collection days. This reduces the effort required to participate and integrates recycling into residents' existing household routines.

A reward system would provide an additional incentive for participation. Households that consistently meet the required recycling participation or quality criteria could receive vouchers or other rewards. This encourages residents not only to recycle regularly, but also to ensure that the materials placed inside the yellow bags are suitable for recycling.

However, convenience and incentives alone may not be sufficient to ensure that residents separate materials correctly. Clear instructions should therefore be provided on the types of materials that can be placed inside the yellow bags. This would help residents understand the requirements and reduce contamination of recyclable materials.

3.2.3 Collection Worker Participation

Collection workers form an important component of the proposed system. Workers would be responsible for collecting filled yellow bags from residents' doorsteps on designated collection days and transporting them to the appropriate recycling facility or sorting location.

The doorstep collection system provides an additional employment opportunity by creating roles related to the collection and handling of recyclable materials. This also removes the need for residents to transport their recyclables themselves, making the system more convenient for households, particularly in high-rise residential buildings.

Collection workers could also perform an initial visual inspection of the yellow bags before collection. Bags containing excessive amounts of general waste or contaminated materials could be identified and flagged for corrective action. This provides an early contamination-control measure before the materials enter the recycling process.

3.2.4 Expected Waste-Reduction and Recycling Mechanism

The main expected effect of the proposed system is to improve the quality and quantity of recyclable materials entering Singapore's recycling stream.

Under the conventional system, recyclable materials may be disposed of together with general household waste or become contaminated before reaching the recycling facility.

Contaminated recyclables may subsequently be rejected during sorting or processing, reducing the amount of material that can actually be recycled.

The proposed mechanism can therefore be represented as:

Dedicated yellow bags → source separation → doorstep collection → contamination checks → higher-quality recyclable materials → improved recycling rate

This directly applies the key lesson identified from Germany's separate collection approach. Instead of allowing recyclable materials to mix with general household waste, the proposed system separates them at the household level before collection.

The recommendation also adapts the German approach to Singapore's high-rise residential environment. Rather than directly replicating Germany's Yellow Bag collection system, the proposed doorstep collection model provides a more convenient method for residents while creating collection roles to support the system.

Importantly, the primary purpose of the proposed solution is to improve recycling efficiency by reducing contamination and the rejection of recyclable materials, rather than simply increasing the amount of waste collected for recycling.

3.2.5 Overall Recommendation

The proposed Yellow Bag Recycling Scheme translates the international benchmarking findings into a practical Singapore-focused intervention by adapting Germany's source-separation approach to Singapore's high-rise residential environment. The scheme combines convenient household recycling, doorstep collection, contamination control and financial incentives to encourage greater participation.

Its objective is not only to increase the amount of recyclable material collected, but also to reduce contamination and prevent recyclable materials from being rejected during the recycling process. By separating clean recyclable materials from general household waste at the source, the scheme aims to improve the quality of recyclable materials entering the recycling stream and ultimately increase Singapore's recycling rate.

The proposed system distributes responsibility across households, collection workers and system operators. Households are responsible for separating suitable recyclable materials, collection workers collect and conduct initial contamination checks, while the wider system manages sorting and recycling. The reward system further encourages households to participate consistently and practise proper waste separation.

Overall, the proposed Yellow Bag Recycling Scheme adapts Germany's successful source-separation approach to Singapore's high-rise environment. By making recycling more convenient through doorstep collection, reducing contamination through collection checks and encouraging participation through rewards, the scheme can improve the quality of recyclable materials and contribute to a higher recycling rate in Singapore.

4. Benchmark Policy Scenario and Potential Impact for recommendation

4.1 Benchmark Scenario 1 : Potential Scale of Improvement.

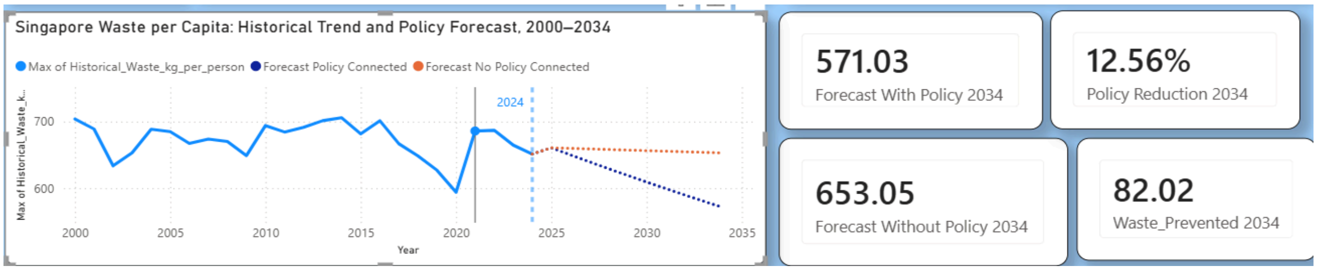
To illustrate the possible scale of improvement, the annualised reduction rates associated with the three selected OECD benchmark countries were used to construct a benchmark policy scenario for Singapore.

The average annual benchmark reduction rate was applied cumulatively to Singapore's existing baseline forecast. Two future trajectories were then compared:

- Forecast Without Policy — continuation of the baseline forecast.
- Forecast With Benchmark Reduction — baseline forecast adjusted by the average annual reduction achieved by the selected benchmark countries.

By 2034, the baseline scenario projects approximately 653.05 kg of waste per person, while the benchmark-adjusted scenario reaches approximately 571.03 kg per person.

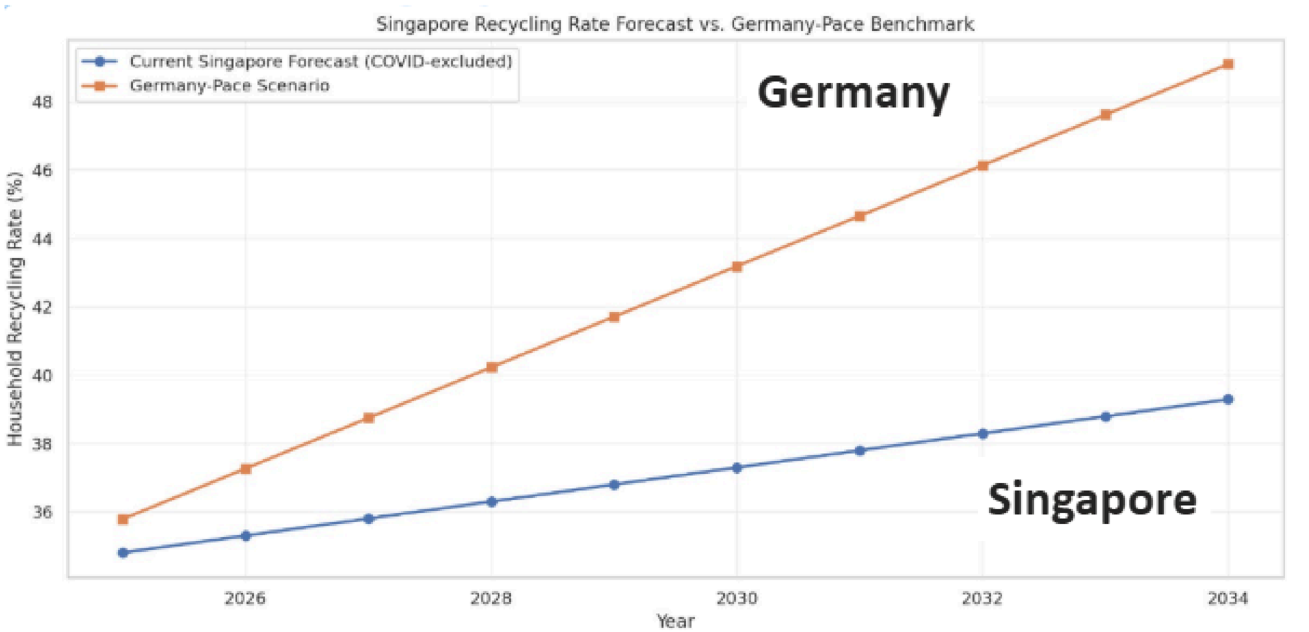
This represents a difference of approximately 82.02 kg per person, equivalent to a 12.56% reduction relative to the baseline forecast.



These results should not be interpreted as a prediction that the Return & Save Scheme alone will produce a 12.56% reduction. The scenario assumes that Singapore is able to achieve a reduction rate comparable to the selected benchmark countries, while actual outcomes would depend on implementation quality, public participation, business adoption and other future economic and policy conditions.

The forecast therefore serves as a scenario-based benchmark of potential improvement, rather than a guaranteed causal estimate of the recommendation's impact.

4.2 Outcome for recommendation 2 (recycling rate)



If Singapore adopts the proposed Germany-inspired source-separation system, its recycling rate is projected to improve at a faster rate compared with the current trend.

Under the Germany-paced scenario, Singapore could achieve an increase of approximately 1.5 percentage points in recycling rate per year, compared with approximately 0.5 percentage points per year under the current trajectory.

This suggests that adopting the proposed solution could accelerate Singapore's recycling improvement and narrow the gap with Germany's recycling performance.

5. Overall Findings & Conclusion

This project's analysis indicates that Singapore faces two closely related but distinct waste management challenges. First, total waste generation continues to increase in absolute terms, even though waste generated per capita has shown a slight decline. Second, recycling rates for several major waste categories remain relatively low, particularly for plastic, and have generally declined since around 2019–2020. Therefore, a combined strategy that focuses on reducing waste at its source while improving the separation and collection of remaining waste could address both challenges simultaneously.

- Strategy 1 — Reduce Waste Generation: Implement the Return & Save takeaway-container scheme, as proposed in Section 3.1, to encourage the reuse of food and beverage containers and reduce the amount of single-use packaging waste generated.
- Strategy 2 — Improve Waste Separation and Collection: Implement the adapted doorstep recycling scheme, as proposed in Section 3.2, to improve the accessibility and convenience of recycling while encouraging households to separate recyclable materials correctly.

Together, these strategies form a Reduce → Separate → Recycle approach aimed at reducing the volume of waste ultimately requiring disposal at Semakau Landfill. By reducing waste generation at the source and improving the recovery of recyclable materials, the proposed approach could help extend the landfill's remaining capacity and contribute to Singapore's broader waste-reduction objectives under the Zero Waste Masterplan.

However, this analysis has several limitations. The findings are based primarily on historical trend analysis and cross-country comparisons rather than controlled experimental evidence. Therefore, the policy comparisons presented in Section 2.5 provide supporting evidence but cannot establish a direct causal relationship between the policies and observed waste-management outcomes. Similarly, the forecast presented in Section 2.3 assumes that historical trends will continue and does not account for potential major disruptions or significant changes in future policies and waste-management practices.

Future research should therefore consider conducting small-scale pilot programmes to evaluate the effectiveness of the proposed strategies in real-world settings. For example, the initiatives could initially be implemented within one HDB town and a selected group of food courts. Key indicators, including participation rates, recycling volumes, contamination rates, and waste reduction levels, should be monitored and evaluated. The results could then be used to refine the proposed strategies and update the forecast once sufficient intervention data become available. This would provide stronger evidence for assessing the feasibility and potential impact of implementing the strategies on a larger scale.

Assumptions & Limitations

- Data Reliability: Data obtained from data.gov.sg and the OECD are assumed to be accurate, consistent, and reliably reported by the respective originating agencies. Any errors or inconsistencies present in the original datasets may therefore affect the results of this analysis.
- Forecasting Assumption: The policy-adjusted 2034 scenario assumes that Singapore could achieve a rate of improvement broadly comparable to the benchmark countries' progress between 2000 and 2020. In contrast, the unadjusted baseline scenario assumes that the historical trends observed in Singapore will continue into the forecast period without significant changes.
- Policy Assumption: The policies identified in Spain, Japan, and the United Kingdom are considered plausible contributing factors to the waste-reduction outcomes observed in these countries. However, these policies are not assumed to be the sole or definitive causes of the observed improvements, as other economic, social, technological, and environmental factors may also have influenced their outcomes.
- Metric Consistency: The recycling rate presented in this project's dashboards is calculated from the domestic waste-category dataset described in Section 1.2. As discussed in Section 2.2.1, this metric should be interpreted as an internally consistent indicator of recycling trends rather than a direct substitute for the National Environment Agency's (NEA) officially published national recycling rate. The two measures differ in terms of their scope, data coverage, and calculation methodology.

References

City of Heidelberg. (n.d.). *The yellow bag*.
https://www.heidelberg.de/english/Home/City+Hall/yellow+bag.html
This is the official Heidelberg municipal source describing the Yellow Bag/Yellow Bin system for separate collection of plastic, metal and composite packaging.

Ministry of Sustainability and the Environment. (2024, August 6). *Written reply to Parliamentary Question on recycling rate*. Government of Singapore.
https://www.mse.gov.sg/latest-news/written-reply-to-pq-on-recycling-rate/
The official page is dated 6 August 2024.

Ministry of Sustainability and the Environment. (2026, May 18). *Waste management*. Government of Singapore.
https://www.mse.gov.sg/policies/waste-management/
The official page lists its last update as 18 May 2026.

National Environment Agency. (n.d.). *Waste statistics and overall recycling*. Government of Singapore. Retrieved August 14, 2026, from

https://www.nea.gov.sg/our-services/waste-management/waste-statistics-and-overall-recycling

National Environment Agency. (2025, July 23). *Waste generation rate continues to trend downwards in 2024*. Government of Singapore.

https://www.nea.gov.sg/media/news/news/index/waste-generation-rate-continues-to-trend-downwards-in-2024

Organisation for Economic Co-operation and Development. (2025). *Environment at a glance indicators*. OECD Publishing.

https://doi.org/10.1787/ac4b8b89-en

Singapore Department of Statistics. (2023). *Indicators on population, annual (2026)* [Data set]. data.gov.sg. Retrieved August 14, 2026, from

https://data.gov.sg/datasets/d_3d227e5d9fdec73f3bcadce671c333a6/view

Singapore Department of Statistics. (2024). *Waste management and overall recycling rates, annual (2026)* [Data set]. data.gov.sg. Retrieved August 14, 2026, from

https://data.gov.sg/datasets/d_daf568968ab40dc81e7b08887a83c8fa/view

Ministerio de Medio Ambiente, y Medio Rural y Marino. (2009). *Resolución de 20 de enero de 2009, por la que se publica el Acuerdo del Consejo de Ministros por el que se aprueba el Plan Nacional Integrado de Residuos para el período 2008–2015*. *Boletín Oficial del Estado*, 49, 19893–20016.

https://boe.es/buscar/doc.php?id=BOE-A-2009-3243

Ministry of the Environment, Government of Japan. (2005, May 26). *Amendment of basic policy for comprehensive and systematic promotion of measures on waste reduction and proper waste disposal*.

https://www.env.go.jp/en/press/2005/0526a.html

WRAP. (2010, September 21). *Evaluation of Courtauld food waste target – Phase 1*.

https://www.wrap.ngo/resources/report/evaluation-courtauld-food-waste-target-phase-1

Team Contributions

Members	Main Contribution
Emmanuel	Data exploration & preparation; Singapore domestic waste generation dashboard (category trends, population, waste per capita) — Section 2.1
Pierre	Singapore recycling-performance dashboard and cause-and-effect insights (overall recycling trend, disposed vs. recycled, plastic recycling) — Section 2.2
Kwan Teng	Waste-per-capita and recycling-rate forecasting dashboard, 2000-2034 — Section 2.3
Zi Qian	OECD waste-reduction benchmarking and policy-scenario dashboard; Return & Save recommendation — Sections 2.4-2.6.1
Hong Jean	Overseas countries' recycling-statistics dashboard (Germany, Switzerland, Sweden); Germany policy analysis and improved waste-separation recommendation — Sections 2.5, 2.6.2